STA 214 Homework 3

Due: Friday, September 19 at 10:00pm on Canvas.

Instructions: There are two parts to this assignment. Part I is practice with maximum likelihood estimation, Part II is practice with logistic regression diagnostics.

Getting started: Begin by downloading the HW 3 template from the course website:

`https://sta214-f25.github.io/homework/hw_03_template.qmd`

Save this template file to your computer, then open it in RStudio. As you complete the assignment, you will write down your answers to all questions in the file, and include all R code in code chunks. *If a question requires code, you will not receive credit if no code is provided.*

Submission: When you have completed the assignment, render your homework to HTML and submit on Canvas. For Part I, you may submit a separate scan of your written work, if you prefer.

Important: you must submit the rendered HTML on Canvas to receive full credit!

1 Maximum likelihood estimation

This part focuses on practice with maximum likelihood estimation. A brief review of mathematical notation and rules for logs and derivatives can be found in the course codebook.

Suppose we have a random variable Y, which can take on values $y = 0, 1, 2, 3, 4, ...$ (i.e., any non-negative integer). The probability of each outcome does not depend on any explanatory variables, and we are told that

$$P(Y = y) = \frac{\lambda^y e^{-\lambda}}{y!}$$

where $\lambda \geq 0$. (Recall that $y!$ is the factorial, with $0! = 1$, and $y! = y(y-1)(y-2)\cdots 1$). So, $P(Y = 0) = e^{-\lambda}$, $P(Y = 1) = \lambda e^{-\lambda}$, etc.

1. We don't know λ, but we get 6 observations:

$$Y_1 = 2,\ Y_2 = 4,\ Y_3 = 6,\ Y_4 = 6,\ Y_5 = 3,\ Y_6 = 1$$

   (a) Write down the likelihood $L(\lambda)$ as a function of λ, using the observed sequence of data.

   (b) Write down the log likelihood $\log L(\lambda)$ as a function of λ, using the observed sequence of data.

   (c) Use calculus to calculate the maximum likelihood estimate $\widehat{\lambda}$ from the observed data. You will:

   - differentiate the log likelihood (hint: the factorials will go away when you differentiate)
   - set the derivative equal to 0
   - solve for $\widehat{\lambda}$

   Make sure to show all your work.

2 Data analysis

Here we work with data from a website called ScienceForums.Net (SFN), which has been open since 2002 and hosts conversations on a range of topics from biological and physical science to religion and philosophy. Each row in the data represents one 'thread', which is comprised of a series of posts stemming from an initial post. For each thread, we have some information that SFN collects such as the number of views and the number of authors. The threads present in the data are a random sample of threads from 2002-2014, with the data collected in 2014. SFN moderators are interested in using this data to determine which threads warrant the most attention.

You can load the SFN data into R by

```
sfn <- read.csv("https://sta214-f25.github.io/homework/sfn.csv")
```

The sfn dataset contains the following columns:

- Age: the age of the thread (in days) when the data was collected in 2014, measured from the first post in the thread
- State: sometimes moderators close threads if they are inappropriate. closed indicates the thread has been closed, otherwise State is open
- Posts: the number of posts in the thread
- Views: the total number of views of the thread
- Duration: the number of days between the first and last posts in the thread
- Authors: the number of distinct authors posting in the thread
- AuthorExperience: the number of days the author of the first post in the thread had been registered on SFN when the thread began (0 indicates they registered that day)
- DeletedPosts: the number of posts in the thread that have been deleted by a moderator
- Forum: the forum in which the thread was posted (e.g., Science)
- AuthorBanned: whether the original author of the thread is currently banned from posting on SFN (at the time of data collection, not when the thread was first posted)

Research question: Suppose you have been approached by moderators at SFN. They give you the data, and ask the following question:

- Is there a relationship between the number of Posts in a thread and whether a thread will have *at least one* deleted post, after accounting for the number of Views, the number of Authors, and the Forum?

3. Here you will begin to use logistic regression to answer the moderators' question.

   (a) Which variables should we focus on to answer the moderators' question? Which of these is our response variable, and which will be our explanatory variables, for logistic regression? *Hint: you may need to create a new variable for the response!*

   (b) Before fitting a model, let's see if any of the quantitative variables need transformations.

- Create empirical logit plots to summarize the relationship between each quantitative predictor and your binary response. See the class activity and the course codebook for details on empirical logit plots.
- Using the empirical logit plots, discuss whether any transformations are needed on the explanatory variables.

(c) Based on your empirical logit plots, write down a logistic regression model that will allow you to answer the moderators' question. Describe how you will use the model to answer their question.

(d) Fit your model from (c), and report the equation of the fitted model.

(e) Assess the shape assumption for your fitted model: create quantile residual plots to check the shape assumption for quantitative variables (you may use the `qresid` function in the `statmod` package). See the class activity and the course codebook for details on quantile residual plots.

(f) If there are any violations of the shape assumption in part (e), experiment with further transformations to address these violations. If you made any changes to your model from (e), report your new fitted model here. If there are no violations of the shape assumption in part (e), simply state this.

(g) Interpret the coefficients of your final logistic regression model in context of the research question.